

Development of a Composite Eutrophication Proxy Index (CEPI) Using Sentinel-2 GeoAI Time-Series for Lake Toba

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ABSTRACT

Lake Toba faces increasing environmental pressure due to eutrophication, characterized by excessive algal growth and declining water quality. Conventional monitoring based on in situ measurements has spatial and temporal limitations, making it difficult to fully capture the spatial and temporal dynamics of eutrophication in large lake environments. Therefore, an efficient and sustainable alternative approach is needed for spatial-temporal water quality assessment. This study proposes the development of a Composite Eutrophication Proxy Index (CEPI) based on Sentinel-2 imagery and GeoAI as a preliminary bio-optical proxy framework for eutrophication-related monitoring under limited field-data conditions. The analysis utilized Sentinel-2 Level-2A imagery from 2021 to 2025 within a 100-meter shoreline buffer zone. The CEPI was constructed by integrating normalized NDCI, AFAI, and TSS components. Temporal analysis, K-Means clustering, Isolation Forest anomaly detection, and statistical evaluation were conducted to assess the spatial-temporal consistency and behavior of the index. CEPI values ranged from 0.30 to 0.75, with an average value of 0.50, indicating relative mesotrophic-like conditions. Seasonal fluctuations were observed, with higher CEPI values generally occurring at the end of the rainy season and the beginning of the dry season. Spatial clustering identified relative CEPI zones concentrated around floating net cage aquaculture areas. Strong correlations were observed between CEPI and its constituent bio-optical components, particularly AFAI ($r = 0.88$) and NDCI ($r = 0.81$). Temporal stability analysis produced a coefficient of variation (CV) of 0.029, indicating relatively stable temporal behavior. Internal consistency assessment yielded R^2 values of 0.997 for training data and 0.930 for temporal test data, indicating temporal coherence within the CEPI framework. The results indicate that CEPI provides a consistent spatial-temporal proxy for eutrophication-related dynamics in Lake Toba and may support scalable monitoring approaches for tropical lakes under data-limited conditions.

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1. INTRODUCTION

Lake Toba is the world's largest volcanic lake and plays a significant role in Indonesia's ecology, economy, and socio-culture. However, its ecosystem is currently facing serious environmental pressures, with eutrophication being one of the primary threats [1]. Eutrophication is the process of excessive nutrient enrichment, particularly nitrogen and phosphorus, in water bodies, which can trigger algal blooms, reduced dissolved oxygen levels, and overall water quality degradation [2]. This condition not only threatens biodiversity but also has the potential to undermine the sustainability of the fisheries and tourism sectors, which serve as the economic backbone of communities around Lake Toba.

Various studies have confirmed the presence of eutrophication symptoms in Lake Toba. Lukman [3], demonstrated that Lake Toba's trophic status varies from oligotrophic to eutrophic, influenced by human activities, sedimentation, and fish feed residues from aquaculture operations. Wojewódka-Przybył et al. [4], also confirm that the primary sources of nutrients stem from floating net cage (FNC) operations, domestic wastewater, and runoff from agriculture and deforestation in the watershed. In July and August 2025, the public reported changes in water color and mass fish deaths at several locations, which were suspected to be caused by a combination of upwelling, organic waste pollution, sediment runoff, and dry season conditions [5–7]. However, the exact cause has not yet been fully determined, necessitating a more comprehensive scientific analysis of water quality dynamics and eutrophication risks.

Conventional eutrophication monitoring methods using direct (in situ) sampling have limitations in large lake areas such as Lake Toba because they are costly, time-consuming, and unable to fully represent spatial and temporal variations. To address these limitations, Sentinel-2 satellite imagery-based remote sensing technology serves as an effective alternative. With high spatial resolution (10 and 20 m) and an acquisition frequency of approximately every five days, Sentinel-2 enables continuous, time-series-based monitoring of water quality parameters [8]. Furthermore, satellite data has proven capable of analyzing the spatial and temporal dynamics of chlorophyll-a concentrations as a primary indicator of algal biomass and its relationship with land-use changes in watersheds [9].

Advances in geospatial artificial intelligence (GeoAI) technology, including machine learning and deep learning, are further enhancing the capabilities of satellite-based water quality analysis [10]. Several studies have successfully estimated chlorophyll-a, Total Suspended Solids (TSS), and other bio-optical parameters with high accuracy [11, 12]. However, most GeoAI approaches are still supervised, requiring in situ measurement data for model training and internal evaluation. This poses a unique challenge in Lake Toba, given logistical constraints, costs, the vast expanse of the water body, and geographical conditions that make extensive and sustainable field data collection difficult.

To address these limitations, this study utilizes the integration of high-resolution remote sensing data, geographic information systems (GIS), and cloud computing platforms such as Google Earth Engine (GEE). This approach enables data-driven spatiotemporal analysis without reliance on field calibration. Since satellite imagery only represents the water surface layer, the analysis area is limited to the shallow water zone around Lake Toba by applying a 100-meter buffer from the shoreline toward the lake. This strategy aims to obtain reflectance data that is more representative of the optical conditions of the surface water.

This research proposes the development of a Composite Eutrophication Proxy Index (CEPI) to represent spatial and temporal eutrophication conditions through the integration of three main bio-optical parameters, namely the Normalized Difference Chlorophyll Index (NDCI), Adjusted Floating Algae Index (AFAI), and Total Suspended Solids (TSS) derived from Sentinel-2 imagery. The CEPI model is designed as a calibration-independent proxy approach for spatial-temporal eutrophication assessment under limited field-data conditions. This approach aims to support rapid, consistent, and sustainable monitoring of relative eutrophication dynamics using remote sensing and GeoAI analysis. In addition to supporting water quality assessment in Lake Toba, the CEPI framework also has the potential to be adapted for other tropical lakes with similar environmental characteristics.

This approach is expected to yield a remote sensing and GeoAI-based eutrophication monitoring system that is calibration-free and easy to operate. In addition to supporting the sustainable management of Lake Toba, the CEPI concept also has the potential to be adapted for other tropical lakes with similar characteristics, such as Lake Maninjau and Lake Singkarak, thereby expanding the application of tropical aquatic ecosystem monitoring in Indonesia.

2. RESEARCH METHOD

2.1. Study Location and Time

The study focused on the waters of Lake Toba, with a buffer zone extending 100 meters from the main shoreline. The analysis was conducted as a monthly time series spanning January 2021 to October 2025. This approach was chosen to capture seasonal patterns and interannual trends, which are essential for detecting eutrophication dynamics.

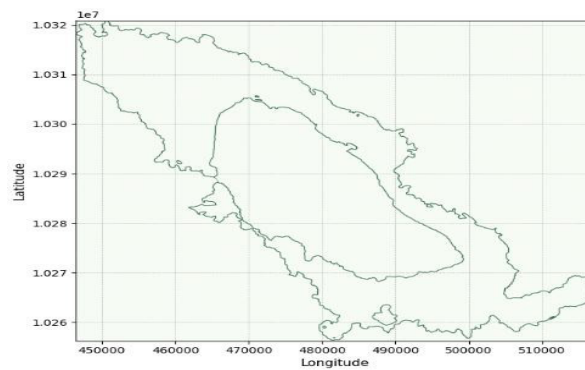


Figure 1. Map of Study Location and 100 m Buffer Zone

2.2. Research Tools and Materials

Sentinel-2 data was used because it has optimal spectral and spatial resolution (20 m) for water quality studies, particularly in the red-edge and near-infrared bands, which are sensitive to chlorophyll content and suspended matter [13].

Table 1. Data Availability

Data Types	Description	Source
Sentinel-2 L2A Satellite Image	Used for spectral index calculations	Google Earth Engine (GEE)
Study Area Boundary & 100 m buffer	Polygon of Lake Toba water area and 100 m buffer	OpenStreetMap [14]
Distribution of KJA	Results of manual digitization of high-resolution imagery and KJA points	Google Maps + QGIS

Table 2. Software used

Software	Main Functions
Google Earth Engine	Acquisition, pre-processing, and spectral index calculation
Python 3.12.5	Spatio-temporal CEPI analysis and application of unsupervised learning algorithms and visualization
QGIS 3.40.9	Map visualization, spatial overlay, and internal evaluation of analysis results

2.3. Research Stages

The research flow is as follows:

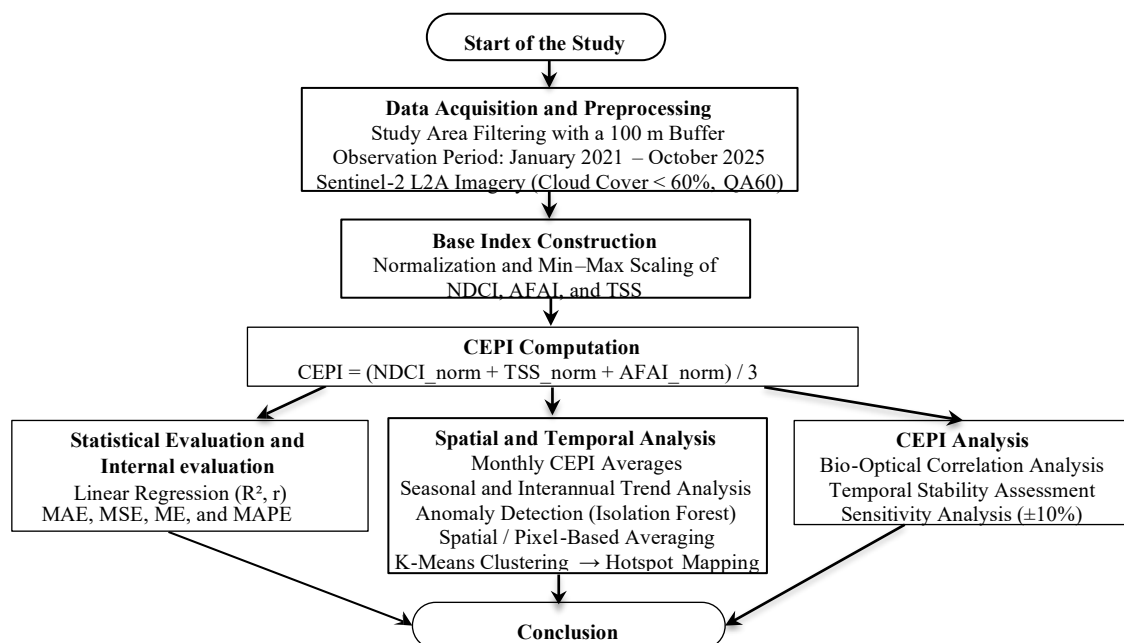


Figure 2. CEPI model development workflow

2.3.1. Data Acquisition and Preprocessing

To ensure the quality and consistency of Sentinel-2 imagery, the following were used:

1. Lake Toba Area Filter (100 m buffer) and the time period from January 2021 to October 2025.
2. Selection of Sentinel-2 Level-2A imagery with 60% cloud cover.
3. Use of the QA60 band to mask cloud and cirrus pixels.

This step is critical because errors in the masking process can affect spectral reflectance values and introduce bias in the estimation of the water optical index [15].

2.3.2. Construction of the Composite Eutrophication Proxy Index (CEPI)

This stage aims to design the CEPI as a composite proxy index that represents the eutrophication potential of various complementary bio-optical signals.

Table 3. Basic Index Calculations

Index	Formula	Reason for Selection
NDCI	$(B5 - B4) / (B5 + B4)$	Selected because it is sensitive to chlorophyll-a content in shallow waters and stable against the effects of turbidity [16, 17]
TSS	$a \times (R(B11)/(1 - R(B11)/c)) + b$	Represents suspended particulate matter and turbidity-related water conditions that influence water quality and can be effectively monitored using Sentinel-2 imagery [18,19].
AFAI	$B8 - [B6 + (B11 - B6) \times (\lambda B8 - \lambda B6) / (\lambda B11 - \lambda B6)]$	Good for detecting the presence of surface algae and more stable than FAI under tropical atmospheric conditions. [20, 21]

All indices are normalized using the Min-Max Normalization method to the range 0–1 to ensure equivalence in value contribution. The CEPI value is calculated using the formula:

$$CEPI = \frac{NDCI_{norm} + TSS_{norm} + AFAI_{norm}}{3}$$

A simple average approach (equal weighting) was adopted as an initial baseline configuration because sufficient in situ calibration data were not available during the study period.

2.3.3. Spatio-Temporal Analysis and CEPI Experiments

2.3.3.1. Temporal Analysis of CEPI

Temporal analysis was conducted to evaluate the dynamics of eutrophication and the stability of the CEPI index in Lake Toba. This analysis includes the calculation of monthly average CEPI values from January 2021 to October 2025 to identify seasonal fluctuations and interannual trends. This approach is used to describe temporal changes in water conditions and capture seasonal patterns related to aquatic environmental dynamics.

Temporal anomaly detection was performed using the Isolation Forest algorithm, which is capable of recognizing extreme deviations in the CEPI time series without requiring training data or specific distribution assumptions [22]. This method was used to identify periods with CEPI values deviating from normal patterns, which may indicate unusual variations in eutrophication-related conditions or changes in surface water characteristics.

2.3.3.2. Spatial Analysis of CEPI

Spatial analysis was conducted to map the distribution of eutrophication levels in Lake Toba. In this stage, the average CEPI value per pixel was calculated over the observation period to spatially represent eutrophication conditions. Next, these CEPI values were grouped using the K-Means clustering algorithm to identify relative spatial groupings of CEPI values associated with different eutrophication tendencies. This clustering approach produced a CEPI hotspot map representing water zones with varying levels of eutrophication risk in Lake Toba [23].

2.3.4. Evaluation and Internal evaluation of the CEPI Model

2.3.4.1. CEPI Analysis

To strengthen the interpretation of the modeling results, three complementary follow-up analyses were conducted, namely: Bio-optical correlation, used to assess the relationship between the parameters

NDCI, AFAI, TSS, and CEPI using Pearson's correlation [24]. Temporal stability was evaluated using the coefficient of variation (CV), which is commonly used to quantify temporal variability in environmental datasets [25]. $\pm 10\%$ sensitivity simulations were performed on each component of CEPI to assess the relative contribution of each parameter to changes in the index value using a fuzzy variable model approach [26].

2.3.4.2. Statistical Evaluation and Internal Consistency Assessment

Statistical evaluation was conducted to assess the internal consistency and temporal behavior of the CEPI framework. The evaluation stages include two main procedures: internal evaluation and temporal internal evaluation. In the internal evaluation, the predicted CEPI values will be compared with observed values in the 80% training data and 20% test data using simple linear regression to obtain the coefficient of determination (R^2) and Pearson correlation (r). Subsequently, temporal internal evaluation will be applied to the test data to assess the temporal consistency of the CEPI framework across different observation periods. The relationship between predicted and observed values will be visualized via a scatter plot of predicted versus observed values to illustrate the model's consistency.

In addition to regression, the evaluation will include calculations of MAE, MSE, ME, and MAPE, which are used to measure absolute error, mean squared error, potential systematic bias, and relative error percentage, respectively. These four metrics are used to assess the stability, robustness, and reliability of CEPI as a bio-optical composite index. With this approach, the CEPI framework can be internally evaluated to examine its spatial-temporal consistency and relative behavior.

2.4. Final Research Output

Research outputs include: (1) multitemporal CEPI spatial distribution maps (2021–2025); (2) temporal trend graphs and anomaly detection using Isolation Forest; (3) K-Means-based eutrophication zoning cluster maps; (4) CEPI stability and sensitivity analysis via CV and $\pm 10\%$ simulation; (5) internal consistency assessment and temporal evaluation of the CEPI framework using linear regression with R^2 and correlation (r) measurements; and (6) calculations of MAE, MSE, ME, and MAPE, which are used to measure absolute error, mean squared error, potential systematic bias, and relative error percentage, respectively.

3. RESULTS AND DISCUSSION

3.1. Results of the CEPI Model Design

The CEPI index was constructed by integrating three main bio-optical parameters estimated from Sentinel-2 data, namely: NDCI as an indicator of phytoplankton productivity, TSS as an indicator of turbidity due to suspended solid particles, and AFAI as an indicator of the presence of surface algae.

Table 4. Bio-optical Parameters Composing CEPI and Their Value Ranges

No	Bio-optical Parameters	Value Range (Min–Max)	Average	Standard Deviation
1	NDCI	0.00 – 1.00	0.3976	0.2252
2	AFAI	0.00 – 1.00	0.6081	0.1670
3	TSS	0.00 – 1.00	0.4999	0.1662
4	CEPI	0.00 – 1.00	0.5019	0.1012

Table 4 shows the results of the integration of the three main bio-optical parameters that make up the CEPI. The highest average value was found in AFAI (0.6081), indicating the dominance of reflectance from surface algae in Lake Toba, while NDCI and TSS showed moderate values of 0.3976 and 0.4999, respectively, representing phytoplankton productivity and water turbidity levels. The average CEPI value of 0.5019 with a standard deviation of 0.1012 indicates relatively elevated eutrophication tendencies within the study area.

3.2. Temporal Analysis

3.2.1. Monthly Temporal Analysis of CEPI

Monitoring of monthly CEPI values from January 2021 to October 2025 reveals a consistent pattern of seasonal fluctuations. CEPI values range from 0.30 to 0.75, with an average of approximately

0.50, suggesting predominantly mesotrophic-like conditions in Lake Toba. Increases in CEPI values generally occur from the end of the rainy season through the start of the dry season (March–June) and at the end of the year (November–December). Conversely, decreases in CEPI values occur during the peak of the rainy season (January–February) and the middle of the dry season (July–September). This pattern indicates that CEPI effectively captures the annual cycle of water productivity.

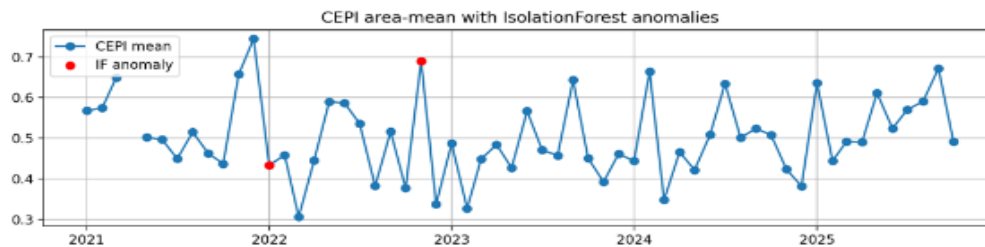


Figure 3. Trends in Average CEPI and Anomaly Detection Using Isolation Forest (2021–2025)

Anomaly detection results using the Isolation Forest (IF) algorithm identified two extreme events during the observation period. A negative anomaly occurred in early 2022, with a drop in the CEPI value to around 0.43, which may indicate temporary changes in surface water conditions or eutrophication-related dynamics. Conversely, a positive anomaly emerged from late 2022 through early 2023, with a spike in the value reaching 0.69. The red dots on the graph mark the months of these anomalies, while the gap around April is due to the absence of valid data caused by heavy cloud cover, making it impossible to calculate the CEPI value for that period.

3.2.2. Temporal Analysis of CEPI in Floating Net Cage (KJA) Hotspot Areas

Analysis of CEPI values by semester at FNC sites in Lake Toba shows dynamic fluctuations during the 2021–2025 period. CEPI values were relatively high in early 2021 (0.55–0.57), dropped sharply in 2022 (0.45), and then rose again to reach a peak of around 0.58 by the end of 2025. This trend may reflect temporal variations in eutrophication-related conditions during 2022. These fluctuations may serve as an early indicator of potential eutrophication.

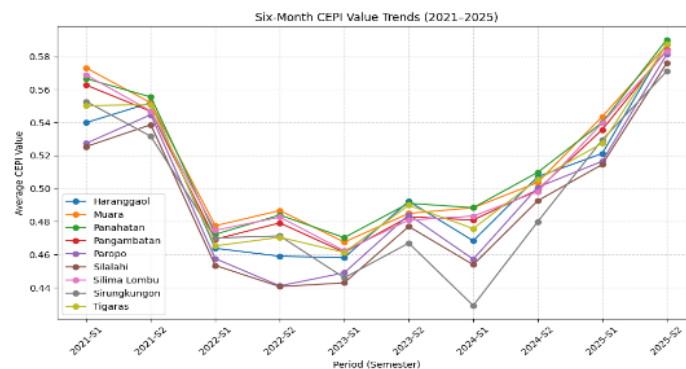


Figure 4. CEPI Time Series and Monthly Anomaly Detection in Hotspot Areas

3.3. Spatial Analysis and GeoAI

3.3.1. Clustering of CEPI Spatial Patterns

Figure 5 displays the spatial classification of the CEPI index using the K-Means algorithm with an optimal number of three clusters ($k = 3$), determined based on the highest Silhouette Score. This configuration is considered the most representative for objectively grouping natural CEPI value patterns without subjective class determination. Spatially, the classification results divide Lake Toba's waters into three relative CEPI zones. The blue zone represents the lowest CEPI ranges and may indicate relatively lower eutrophication tendencies compared to other zones. The yellow zone depicts mesotrophic conditions as a transitional zone with a fairly wide distribution in the central waters of the lake.

Meanwhile, the red zone represents relatively higher CEPI values concentrated around areas with stronger anthropogenic pressure. Overall, the CEPI classification map demonstrates that the unsupervised learning approach is capable of identifying relative spatial patterns of CEPI distribution in Lake Toba. CEPI Clustering of Lake Toba Based on Silhouette Score.

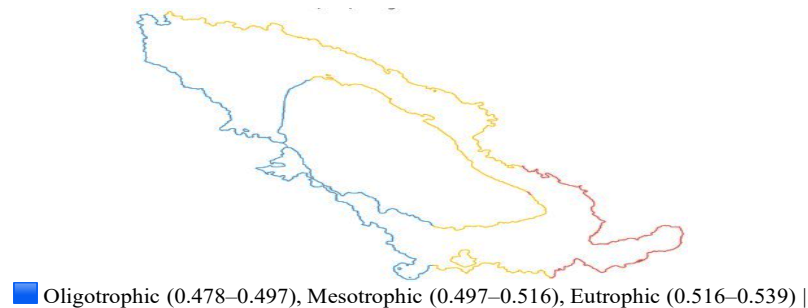


Figure 5. Map of CEPI Clustering Results

Table 5. CEPI Cluster Statistics in Lake Toba using the K-Means algorithm with the optimal number of clusters ($K = 3$) based on the highest Silhouette Score.

Class	Number of Pixels	Area (km ²)	CEPI Min	CEPI Max	Eutrophication Category
1	63,558	25.42	0.4783	0.4967	Oligotrophic–mesotrophic
2	70,633	28.25	0.4967	0.5157	Mesotrophic
3	27,319	10.93	0.5157	0.5386	Mesotrophic–eutrophic

The results of spatial clustering of the CEPI index using the K-Means algorithm with the optimal number of clusters ($K = 3$) indicate a clear differentiation in eutrophication levels in the waters of Lake Toba. The oligo- mesotrophic class dominates the water area with an extent of approximately 25.42 km² and CEPI values ranging from 0.478 to 0.497, reflecting water conditions with relatively low to moderate fertility levels. The mesotrophic class is the largest cluster (28.25 km²) with a CEPI value range of 0.497–0.516, indicating moderate and relatively stable water productivity. Meanwhile, the relative mesotrophic-eutrophic-like conditions has the smallest area (10.93 km²) but exhibits the highest CEPI values (0.516–0.539), indicating a zone with stronger nutrient pressure and the potential to become an eutrophication hotspot. This spatial pattern confirms that ecological pressure in Lake Toba is not evenly distributed.

3.3.2. Relationship Between CEPI and KJA Locations

Table 6. Average CEPI Values at KJA Locations

No	Location	Average CEPI Value
1	Silalahi	0.532
2	Paropo	0.541
3	Muara	0.589
4	Haranggaol	0.562
5	Tigaras	0.570
6	Panahatan	0.589
7	Sirungkunon	0.552
8	Silima Lumbu	0.580
9	Pangambatan	0.576

Table 6 shows the average CEPI values at nine KJA locations in Lake Toba, ranging from 0.53 to 0.59, reflecting mesotrophic to near-mesotrophic–eutrophic conditions. The Muara and Panahatan sites have the highest values (0.589), indicating the strongest nutrient pressure. Conversely, Silalahi recorded the lowest value (0.532), indicating relatively more stable water conditions. The other locations; Paropo, Haranggaol, Tigaras, Sirungkunon, Silima Lumbu, and Pangambatan fall within the intermediate range, reflecting a fairly uniform level of fertility. Overall, the relatively small variation in values suggests that KJA activities exert a uniform influence on the increase in eutrophication across the study area.

3.4. Advanced CEPI Analysis

To strengthen the interpretation of the main results, three further analyses were conducted, namely (1) bio-optical correlation, (2) temporal stability, and (3) CEPI sensitivity experiments.

3.4.1. Bio-Optical Correlation

Pearson correlation analysis was used to assess the relationship between the NDCI, AFAI, TSS, and CEPI parameters [21].

Table 7. Interpretation of the Pearson Correlation Matrix between Bio-optical Parameters and CEPI (2021–2025)

Parameter	NDCI	AFAI	TSS	CEPI
NDCI	1.000	0.761	-0.633	0.814
AFAI	0.761	1.000	-0.432	0.879
TSS	-0.633	-0.432	1.000	-0.160
CEPI	0.814	0.879	-0.160	1.000

Based on the Pearson correlation analysis, CEPI shows strong relationships with AFAI ($r = 0.88$) and NDCI ($r = 0.81$), indicating that the index is strongly influenced by its constituent bio-optical components related to algal presence and chlorophyll dynamics. TSS shows a weak negative correlation with CEPI ($r = -0.16$), meaning that an increase in suspended particles tends to lower the index value due to reduced light penetration and photosynthetic efficiency. Although its contribution is relatively small, the presence of TSS remains important in CEPI as it reflects physical aspects that influence the bio-optical balance of the water. Thus, CEPI integrates biological and turbidity-related optical signals into a single composite framework for relative spatial-temporal eutrophication assessment.

3.4.2. Temporal Stability (Coefficient of Variation)

Temporal stability was evaluated using the coefficient of variation (CV) to measure significant differences between indices [22]. The internal validation results are presented in Table 8 below.

Table 8. Comparison of CEPI Performance with Single Bio-Optical Indices

Evaluation Parameters	NDCI	AFAI	TSS	CEPI
Value Range	0.38–0.45	0.58–0.67	0.47–0.52	0.48–0.54
Average	0.405	0.619	0.500	0.502
Standard Deviation	0.0167	0.0211	0.0134	0.0146
Coefficient of Variation (CV)	0.0413	0.0341	0.0267	0.0290
Correlation with CEPI (r)	0.786	0.997	0.765	1.000
Coefficient of Determination (R^2)	0.619	0.994	0.584	1.000

Internal evaluation results show that CEPI has the highest temporal stability with a CV of 0.029, lower than that of NDCI and AFAI. The very strong correlation between CEPI and AFAI ($r = 0.997$; $R^2 = 0.994$) suggests that AFAI contributes substantially to the CEPI structure because it represents one of the principal bio-optical components within the framework. Meanwhile, NDCI and TSS contribute to variations in chlorophyll and water turbidity.

Figure 6 shows that CEPI has the lowest CV value compared to NDCI, AFAI, and TSS. This indicates that CEPI is the index with the highest temporal stability throughout the 2021–2025 period. The narrow CEPI boxplot distribution indicates minimal inter-temporal variation. NDCI exhibits the widest distribution, indicating significant temporal fluctuations due to the dynamic influence of chlorophyll-a.

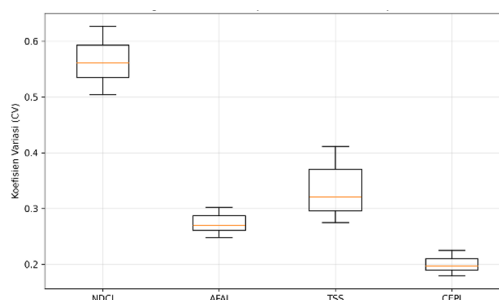


Figure 6. Boxplot Comparison of CV for NDCI, AFAI, TSS, and CEPI

3.4.3. CEPI Sensitivity Experiment

The results of the $\pm 10\%$ sensitivity simulation for the CEPI components (AFAI, NDCI, and TSS) using a fuzzy variable model [23], show that the average CEPI value is 0.497 with a standard deviation of 0.033. This indicates that changes in input parameter values do not cause significant fluctuations in the index. CEPI exhibited relatively stable behavior under $\pm 10\%$ perturbations of its input parameters, indicating that the framework is not excessively sensitive to moderate variations in the constituent bio-optical indices. The AFAI and TSS components have a relatively larger contribution compared to NDCI, confirming the dominant role of algae and suspended solids in water conditions, reinforcing the reliability of CEPI as a remote-sensing-based proxy framework for relative eutrophication assessment under limited field-data conditions.

Table 9. Results of $\pm 10\%$ Sensitivity Simulation

No	Component	Mean	Standard Deviation (SD)
1	AFAI	0.610388	0.169743
2	NDCI	0.398627	0.219654
3	TSS	0.501202	0.163219
4	CEPI	0.497236	0.032684

3.5. Internal Validation and Statistical Performance

3.5.1. Internal evaluation using R^2 and r

Figure 7 displays the relationship between CEPI values derived from the training dataset (80% of the observations) and the corresponding CEPI framework outputs. The analysis yielded $R^2 = 0.997$ and $r = 0.999$, indicating a very strong internal consistency among the variables integrated within the CEPI framework. Figure 8 presents the temporal internal evaluation using the hold-out dataset (20% of the observations), yielding $R^2 = 0.930$ and $r = 0.965$. These results indicate that the CEPI framework exhibits consistent temporal behavior across different observation periods. The reported R^2 and r values are intended solely to describe internal consistency and should not be interpreted as evidence of external validation or predictive accuracy.

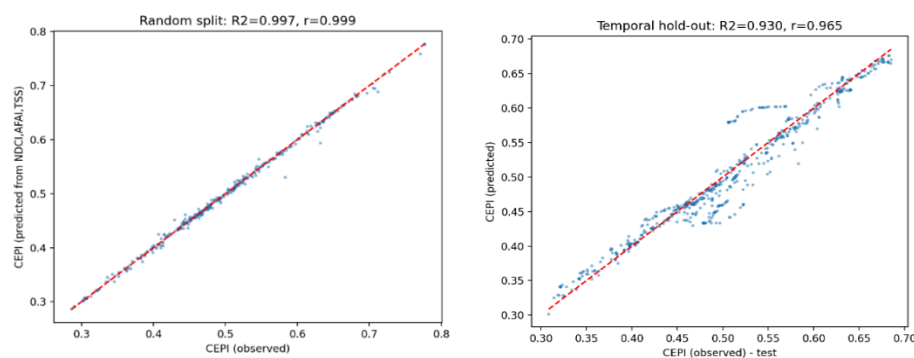


Figure 7 CEPI Evaluation: Predicted vs. Observed Results, Figure 8. CEPI Internal evaluation on Temporal Test Data

3.5.2. Internal evaluation Using MAE, MSE, ME, and MAPE

Table 10. Statistical Evaluation Using MAE, MSE, ME, and MAPE

Evaluation Methods	Value	Interpretation
MAE (Mean Absolute Error)	0.0791	Indicates that the average absolute deviation of the monthly CEPI from the mean CEPI is relatively small
MSE (Mean Squared Error)	0.01	Low squared error, indicating the spatial-temporal stability of the CEPI
ME (Mean Error)	0.0016	Value close to zero, indicating the absence of systematic bias (over/underestimation)
MAPE (Mean Absolute Percentage Error)	15.78	Internal relative variation is still within acceptable limits for satellite-based environmental indices

In addition to internal evaluation using R^2 and r , four other evaluation metrics were used: Mean Absolute Error (MAE), Mean Squared Error (MSE), Mean Error (ME), and Mean Absolute Percentage Error (MAPE). The evaluation results show that the average MAE value is 0.079 and the average MSE value is 0.010, indicating that the deviation of the monthly CEPI values from the average CEPI value is relatively small. This indicates relatively stable spatial-temporal behavior of CEPI within the analyzed dataset. The average ME value of 0.0016 indicates that the difference between the monthly CEPI and the CEPI mean is not systematic, so the CEPI index does not tend to exhibit overestimation or underestimation bias. Thus, CEPI_{mean} may provide a relative spatial representation of average eutrophication-related conditions per pixel. The average MAPE value of 15.78% indicates the presence of relative variation across temporal bands. The MAPE value, which does not approach zero, is primarily influenced by bands with low CEPI values, which mathematically increase the error percentage. In the context of this study, MAPE is not interpreted as a measure of prediction accuracy but rather as an indicator of internal variation and temporal heterogeneity of the CEPI. Therefore, the obtained MAPE values are still acceptable and reflect the natural bio-optical dynamics of Lake Toba's waters. Overall, the statistical assessment indicates that CEPI demonstrates relatively stable and spatially-temporally consistent behavior within the study framework.

3.6. Implications for Lake Toba Management

The findings of this study indicate that the CEPI model can characterize the eutrophication conditions of Lake Toba both spatially and temporally. Integrating CEPI with GeoAI (K-Means) and an anomaly detection algorithm (Isolation Forest) has the potential to support the development of spatial-temporal monitoring frameworks for eutrophication-related conditions based on Sentinel-2 imagery. This approach offers methodological novelty by combining three key aspects:

1. Development of a calibration-free composite index (CEPI),
2. Integration of GeoAI spatial-temporal analysis,
3. Performance evaluation through sensitivity and temporal stability analysis.

3.7. Study Limitations

This study has several limitations that need to be considered in interpreting the results. First, the CEPI framework was developed without direct in situ measurements such as chlorophyll-a concentration, nutrient data, or Secchi depth observations. Therefore, the present study focuses on evaluating the internal consistency and spatial-temporal characteristics of the index rather than establishing direct equivalence with conventional limnological trophic-state assessment. Second, the equal-weighting scheme applied in the construction of CEPI was adopted as an initial baseline approach due to the limited availability of field calibration data. In practice, the contribution of each bio-optical parameter may vary depending on seasonal conditions, sediment dynamics, and local environmental factors. Further studies integrating field observations, limnological measurements, and hydrometeorological data are recommended to improve the ecological interpretation, calibration capability, and broader applicability of the CEPI framework. Further studies integrating field observations and hydrometeorological data are recommended to improve the ecological interpretation and broader applicability of the CEPI framework. The present study does not aim to replace conventional limnological trophic-state assessment. Instead, CEPI is proposed as a scalable remote-sensing-based proxy framework for identifying relative spatial-temporal eutrophication-related patterns, particularly in regions with limited in situ monitoring availability.

4. CONCLUSION

This study developed and applied the Composite Eutrophication Proxy Index (CEPI) framework as a Sentinel-2-based bio-optical proxy approach for the spatial-temporal assessment of eutrophication-related conditions in Lake Toba. By integrating NDCI, AFAI, and TSS, the CEPI framework was used to represent relative bio-optical variations associated with eutrophication dynamics under limited field-data conditions.

The analysis results showed that CEPI values ranged from 0.30 to 0.75, with an average value of 0.50 indicating relative mesotrophic-like conditions. Temporal analysis revealed seasonal fluctuations in

CEPI distribution, while spatial analysis indicated relatively higher CEPI zones around areas with dense Floating Net Cage (FNC) activity. Internal consistency evaluation indicated relatively strong consistency within the analyzed dataset, with R^2 values of 0.997 for the training data and 0.930 for the temporal test data. In addition, the coefficient of variation (CV) value of 0.029 indicated relatively stable temporal behavior during the observation period. Sensitivity experiments at $\pm 10\%$ also showed that CEPI remained responsive yet relatively stable under variations in input parameters.

Overall, the findings indicate that CEPI may support relative spatial-temporal assessment of eutrophication-related conditions in Lake Toba using Sentinel-2 imagery and GeoAI analysis, while providing a complementary remote-sensing-based approach for monitoring under data-limited conditions. The results also indicate the potential of satellite-based and GeoAI-assisted methods for supporting scalable water quality assessment in large tropical lakes.

Future studies are recommended to integrate CEPI with field observations such as chlorophyll-a concentration, nutrient measurements, and Secchi depth data to improve ecological interpretation and calibration capability. The incorporation of hydrometeorological variables, including rainfall, temperature, and water flow, may also provide a better understanding of the environmental factors influencing eutrophication dynamics in Lake Toba. In addition, the development of an online spatial-temporal monitoring platform could support real-time access to CEPI-based analysis for researchers, policymakers, and local communities. Further applications in other tropical lakes, such as Lake Maninjau and Lake Singkarak, are also needed to evaluate the transferability of the proposed CEPI framework under different geographical and environmental conditions.

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CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest regarding the publication of this paper.

DATA AVAILABILITY

The data associated with this study are available from the corresponding author upon reasonable request. Sentinel-2 satellite imagery used in this study is publicly available through the Copernicus Data Space Ecosystem.

AUTHOR CONTRIBUTIONS

Hardinan Sinaga: Conceptualization; Data Curation; Investigation; Methodology; Software; Validation; Formal Analysis; Visualization; Writing-Original Draft. **Kristoko Dwi Hartomo:** Conceptualization; Methodology; Supervision; Validation; Writing-Review & Editing. **Sri Yulianto Joko Prasetyo:** Conceptualization; Methodology; Supervision; Validation; Writing-Review & Editing.

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